

LECTURE 11

GMM I

Definition

Suppose that an econometrician observes data $\{W_i : i = 1, \dots, n\}$, where W_i is a random p -vector. Let g be an l -dimensional function depending on W_i and the k -vector of parameters b :

$$g(W_i, b) = \begin{pmatrix} g_1(W_i, b) \\ \vdots \\ g_l(W_i, b) \end{pmatrix},$$

and $g_j : \mathbb{R}^p \times \mathbb{R}^k \rightarrow \mathbb{R}$ for $j = 1, \dots, l$. The model is defined by the following *moment condition*:

$$\mathbb{E}[g(W_i, \beta)] = 0 \text{ for some } \beta \in \mathbb{R}^k. \quad (1)$$

Examples:

- **Linear regression.** Let $W_i = (Y_i, X_i^\top)^\top$ and $Y_i = X_i^\top \beta + U_i$, where $\beta \in \mathbb{R}^k$, and $\mathbb{E}[X_i U_i] = 0$. In this case, $g(W_i, b) = X_i (Y_i - X_i^\top b)$, $l = k$, and the moment condition is $\mathbb{E}[X_i (Y_i - X_i^\top \beta)] = 0$.
- **IV regression.** Let $W_i = (Y_i, X_i^\top, Z_i^\top)^\top$, $Y_i = X_i^\top \beta + U_i$, where $\beta \in \mathbb{R}^k$, and $\mathbb{E}[Z_i U_i] = 0$, where Z_i is an l -vector. In this case, $g(W_i, b) = Z_i (Y_i - X_i^\top b)$ with the moment condition $\mathbb{E}[Z_i (Y_i - X_i^\top \beta)] = 0$.
- **Lucas's model.** (This example uses time subscripts t rather than i , reflecting the time-series nature of the data.) Suppose that in period t , investors derive utility from consumption C_t . Let $R_{j,t}$ be the rate of return on risky asset j . Suppose that there are m assets. Assume that the utility function is $\sum_{t=1}^{\infty} \delta^t C_t^{1-\alpha} / (1-\alpha)$. In equilibrium, the returns on risky assets are determined by the following Euler equations:

$$\mathbb{E} \left[\delta \left(\frac{C_{t+1}}{C_t} \right)^{-\alpha} (1 + R_{j,t+1}) \right] = 1, \quad j = 1, \dots, m.$$

Here $W_t = (C_t, R_{1,t}, \dots, R_{m,t})$, $b = (a, d)$, $g_j(W_t, b) = d \left(\frac{C_{t+1}}{C_t} \right)^{-\alpha} (1 + R_{j,t+1}) - 1$ for $j = 1, \dots, m$, and the moment conditions are given by the above equations. Here, g is nonlinear in the parameters.

The model is *identified* if $\mathbb{E}[g(W_i, \beta)] = 0$ and $\mathbb{E}[g(W_i, \tilde{\beta})] = 0$ imply $\beta = \tilde{\beta}$, i.e., the solution of (1) is unique. The moment condition gives us l restrictions for k parameters. A necessary condition for the model to be identified is that $l \geq k$, i.e., there must be at least k restrictions. This necessary condition is called the *order condition*. The model is *underidentified* if the order condition fails.

When $k = l$, applying the MM principle, we can estimate β by the value of b that solves the sample analogue of (1):

$$n^{-1} \sum_{i=1}^n g(W_i, \hat{\beta}_n^{MM}) = 0.$$

When $l > k$, no $b \in \mathbb{R}^k$ generally solves all l equations exactly. In this case, we can choose the value of b that makes the sample moments as close to zero as possible. Let A_n be a (possibly random) $l \times l$ weight matrix such that $A_n \rightarrow_p A$, where A is non-random and has full rank l . The *Generalized Method of Moments (GMM) estimator* of β is defined to be the value of b that minimizes the weighted distance of $n^{-1} \sum_{i=1}^n g(W_i, b)$ from zero:

$$\begin{aligned} \hat{\beta}_n^{GMM} &= \arg \min_{b \in B} \left\| A_n n^{-1} \sum_{i=1}^n g(W_i, b) \right\|^2 \\ &= \arg \min_{b \in B} \left(n^{-1} \sum_{i=1}^n g(W_i, b) \right)^\top A_n^\top A_n \left(n^{-1} \sum_{i=1}^n g(W_i, b) \right). \end{aligned} \quad (2)$$

The set $B \subset \mathbb{R}^k$ is assumed to be compact. The matrix $A^\top A$ is positive definite.

Linear case

In this section, we discuss the IV regression example in detail. Here g is linear in the parameters. As in Lecture 10, we assume that some or all of the k regressors in X_i are endogenous:

$$\mathbb{E}[X_i U_i] \neq 0,$$

and that the l instruments Z_i are weakly exogenous:

$$\mathbb{E}[Z_i U_i] = 0.$$

The model is identified if the following *rank condition* is satisfied:

$$\text{rank}(\mathbb{E}[Z_i X_i^\top]) = k.$$

If the rank condition is satisfied and $l = k$, we say that the model is *exactly* or *just* identified. We say that the model is *overidentified* if the rank condition is satisfied and $l > k$ (there are more instruments than parameters). Unlike in Lecture 10, the model may be overidentified here.

In the linear IV regression case, $\hat{\beta}_n^{GMM}$ is the minimizer of

$$\left(n^{-1} \sum_{i=1}^n Z_i (Y_i - X_i^\top b) \right)^\top A_n^\top A_n \left(n^{-1} \sum_{i=1}^n Z_i (Y_i - X_i^\top b) \right),$$

and is given by the following expression:

$$\hat{\beta}_n^{GMM} = \left(\sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) \sum_{i=1}^n Z_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) \sum_{i=1}^n Z_i Y_i.$$

We show next that the GMM estimator is consistent. We need the following assumptions.

- $\{(Y_i, X_i, Z_i) : i = 1, \dots, n\}$ are iid.
- $Y_i = X_i^\top \beta + U_i$, where $\beta \in \mathbb{R}^k$.
- $\mathbb{E}[Z_i U_i] = 0$.
- $\mathbb{E}[Z_i X_i^\top]$ has rank k .
- $A_n \rightarrow_p A$, where A has rank $l \geq k$.
- $\mathbb{E}[X_{i,j}^2] < \infty$ for all $j = 1, \dots, k$.
- $\mathbb{E}[Z_{i,j}^2] < \infty$ for all $j = 1, \dots, l$.

Write

$$\hat{\beta}_n^{GMM} = \beta + \left(n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i U_i.$$

The last two assumptions imply that

$$\mathbb{E}|X_{i,r} Z_{i,s}| < \infty \quad \text{for all } r = 1, \dots, k \text{ and } s = 1, \dots, l.$$

By the WLLN,

$$n^{-1} \sum_{i=1}^n X_i Z_i^\top \rightarrow_p E[X_i Z_i^\top].$$

Since $A_n \rightarrow_p A$, we also have that

$$n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i X_i^\top \rightarrow_p E[X_i Z_i^\top] (A^\top A) E[Z_i X_i^\top].$$

Further, since $E[Z_i X_i^\top]$ has rank k and A has rank $l \geq k$, it follows that the $k \times k$ matrix $E[X_i Z_i^\top] (A^\top A) E[Z_i X_i^\top]$ has full rank k and is therefore invertible. Consequently, by Slutsky's Theorem,

$$\left(n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} \rightarrow_p (E[X_i Z_i^\top] (A^\top A) E[Z_i X_i^\top])^{-1}.$$

Next, by the WLLN,

$$n^{-1} \sum_{i=1}^n Z_i U_i \rightarrow_p 0,$$

and thus $\hat{\beta}_n^{GMM} \rightarrow_p \beta$.

To show asymptotic normality, we will need the following three assumptions in addition to the above.

- $E[Z_{i,j}^4] < \infty$ for all $j = 1, \dots, l$.
- $E[U_i^4] < \infty$.
- $E[U_i^2 Z_i Z_i^\top]$ is positive definite.

Write

$$n^{1/2} (\hat{\beta}_n^{GMM} - \beta) = \left(n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1/2} \sum_{i=1}^n Z_i U_i.$$

The last two assumptions imply that the variance of $Z_i U_i$, $E[U_i^2 Z_i Z_i^\top]$, is finite. By the CLT,

$$n^{-1/2} \sum_{i=1}^n Z_i U_i \rightarrow_d N(0, E[U_i^2 Z_i Z_i^\top]).$$

Define

$$\begin{aligned} Q &= E[Z_i X_i^\top], \\ \Omega &= E[U_i^2 Z_i Z_i^\top]. \end{aligned}$$

Combining the above results, we have

$$n^{1/2} (\hat{\beta}_n^{GMM} - \beta) \rightarrow_d N(0, V),$$

where V takes the sandwich form:

$$V = (Q^\top A^\top A Q)^{-1} Q^\top A^\top A \Omega A^\top A Q (Q^\top A^\top A Q)^{-1}.$$

The variance-covariance matrix V can be estimated by replacing A , Q , and Ω with their consistent estimators A_n , $\hat{Q}_n$, and $\hat{\Omega}_n$, respectively, where

$$\begin{aligned} \hat{Q}_n &= n^{-1} \sum_{i=1}^n Z_i X_i^\top, \\ \hat{\Omega}_n &= n^{-1} \sum_{i=1}^n \hat{U}_i^2 Z_i Z_i^\top, \end{aligned}$$

and $\hat{U}_i = Y_i - X_i^\top \hat{\beta}_n^{GMM}$.

General case

In the general case, the GMM estimator minimizes the nonlinear function in (2). Usually, we do not have a closed-form expression for $\hat{\beta}_n^{GMM}$, and the minimization must be done using numerical procedures. Nevertheless, under general regularity conditions, it is possible to show that $\hat{\beta}_n^{GMM}$ is consistent and asymptotically normal. We provide heuristic arguments for consistency and asymptotic normality.

Since the criterion function in (2) involves averages, we should expect that

$$\left\| A_n n^{-1} \sum_{i=1}^n g(W_i, b) \right\|^2 \rightarrow_p \|A E[g(W_i, b)]\|^2. \quad (3)$$

Assuming that the model is uniquely identified, $E[g(W_i, b)] = 0$ if and only if $b = \beta$. Since $\|A E[g(W_i, b)]\|^2 > 0$ for all $b \neq \beta$, the true value β is the unique minimizer of $\|A E[g(W_i, b)]\|^2$. Intuitively, $\hat{\beta}_n^{GMM}$ is consistent because

$$\begin{aligned} \hat{\beta}_n^{GMM} &= \arg \min_{b \in B} \left\| A_n n^{-1} \sum_{i=1}^n g(W_i, b) \right\|^2 \\ &\rightarrow_p \arg \min_{b \in B} \|A E[g(W_i, b)]\|^2 \\ &= \beta. \end{aligned}$$

The formal proof of consistency requires regularity conditions: (i) compactness of B , (ii) continuity of $g(w, b)$ in b for each w , (iii) *uniform* convergence of the criterion over $b \in B$ in (3), and (iv) unique identification, i.e., β is the unique minimizer of the population criterion.

For asymptotic normality, $\hat{\beta}_n^{GMM}$ solves the first-order conditions:

$$\left(n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \hat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n n^{-1} \sum_{i=1}^n g(W_i, \hat{\beta}_n^{GMM}) = 0. \quad (4)$$

In fact, it is sufficient if $\hat{\beta}_n^{GMM}$ solves the first-order conditions approximately: the right-hand side of (4) may be $o_p(n^{-1/2})$ rather than exactly zero. This accommodates numerical optimization where the solution is found up to a tolerance that shrinks faster than $n^{-1/2}$.

Next, using the expansion of $g(W_i, \hat{\beta}_n^{GMM})$ around $g(W_i, \beta)$ (the element-by-element mean value theorem), we obtain

$$g(W_i, \hat{\beta}_n^{GMM}) = g(W_i, \beta) + \frac{\partial g(W_i, \hat{\beta}_n^*)}{\partial b^\top} (\hat{\beta}_n^{GMM} - \beta), \quad (5)$$

where $\hat{\beta}_n^*$ is between $\hat{\beta}_n^{GMM}$ and β . Substitution of (5) into (4) gives

$$\begin{aligned} 0 &= \left(n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \hat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n n^{-1} \sum_{i=1}^n g(W_i, \beta) \\ &\quad + \left(n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \hat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n \left(n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \hat{\beta}_n^*)}{\partial b^\top} \right) (\hat{\beta}_n^{GMM} - \beta). \end{aligned}$$

We can write

$$\begin{aligned}
& n^{1/2} \left(\widehat{\beta}_n^{GMM} - \beta \right) \\
&= - \left(\left(n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n \left(n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^*)}{\partial b^\top} \right) \right)^{-1} \\
&\quad \times \left(n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n n^{-1/2} \sum_{i=1}^n g(W_i, \beta).
\end{aligned}$$

Since $E[g(W_i, \beta)] = 0$, under regularity conditions,

$$n^{-1/2} \sum_{i=1}^n g(W_i, \beta) \rightarrow_d N \left(0, E[g(W_i, \beta) g(W_i, \beta)^\top] \right).$$

(The asymptotic variance depends on the unknown β .) Since $\widehat{\beta}_n^{GMM}$ is consistent, and, as a result, $\widehat{\beta}_n^* \rightarrow_p \beta$ as well, under regularity conditions,

$$\begin{aligned}
n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^{GMM})}{\partial b^\top} &\rightarrow_p E \left[\frac{\partial g(W_i, \beta)}{\partial b^\top} \right], \\
n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^*)}{\partial b^\top} &\rightarrow_p E \left[\frac{\partial g(W_i, \beta)}{\partial b^\top} \right],
\end{aligned}$$

and that the matrix

$$\left(E \left[\frac{\partial g(W_i, \beta)}{\partial b^\top} \right] \right)^\top A^\top A \left(E \left[\frac{\partial g(W_i, \beta)}{\partial b^\top} \right] \right)$$

is invertible. Then,

$$n^{1/2} \left(\widehat{\beta}_n^{GMM} - \beta \right) \rightarrow_d N(0, V),$$

where

$$\begin{aligned}
V &= (Q^\top A^\top A Q)^{-1} Q^\top A^\top A \Omega A^\top A Q (Q^\top A^\top A Q)^{-1}, \\
Q &= E \left[\frac{\partial g(W_i, \beta)}{\partial b^\top} \right], \\
\Omega &= E[g(W_i, \beta) g(W_i, \beta)^\top].
\end{aligned}$$

The variance-covariance matrix V can be estimated by replacing A , Q , and Ω with their consistent estimators A_n , $\widehat{Q}_n$, and $\widehat{\Omega}_n$, respectively, where

$$\begin{aligned}
\widehat{Q}_n &= n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^{GMM})}{\partial b^\top}, \\
\widehat{\Omega}_n &= n^{-1} \sum_{i=1}^n g(W_i, \widehat{\beta}_n^{GMM}) g(W_i, \widehat{\beta}_n^{GMM})^\top.
\end{aligned}$$